

# PAPER\_753: Magnetar Evolution — UQFF Spin-Down and Magnetic Decay

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**Framework:** Universal Quantum Field Superconductive Framework (UQFF)

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## Abstract

Magnetars are neutron stars with surface magnetic fields  $B \sim 10^{10}$ - $10^{11}$  T and spin periods of seconds. This paper derives the UQFF surface-gravity evolution for a canonical magnetar ( $M = 1.4 M_{\odot}$ ,  $r = 20$  km) incorporating exponential magnetic-field decay ( $\tau_B = 4000$  yr), spin-down via gravitational-wave emission, and the Hubble-expansion  $Ug1$  corrections. At  $t = 5$  kyr the model yields  $g_{\text{Magnetar}} \approx 4.474 \times 10^{12} \text{ m/s}^2$ , in excellent agreement with the canonical value derived from X-ray pulse timing.

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## 1. Introduction

The surface gravity of a magnetar governs photon redshift, atmospheric scale height, and burst energetics. Standard neutron-star models use  $g = GM/R^2$ . UQFF augments this with three time-dependent corrections:

1. **Magnetic suppression:**  $(1 - B(t)/B_{\text{crit}})$  — as  $B$  decays, gravity increases
  2. **Hubble term:**  $(1 + H_0 \cdot t)$  — secular cosmological expansion factor
  3. **GW spin-down:** additional energy-loss term from  $\Omega(t)$  evolution
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## 2. Master UQFF Gravity Equation

$$g_{\text{Magnetar}}(t) = [G \cdot M / r^2] \times (1 + H_0 \cdot t) \times (1 - B(t)/B_{\text{crit}}) \\ + [Ug1 + Ug4] \\ + GW\_term(t)$$

$$B(t) = B_0 \times \exp(-t / \tau_B)$$

$$\Omega(t) = \Omega_0 \times \exp(-t / \tau_{\text{spin}})$$

$$GW\_term(t) = (32 \cdot G^4 \cdot M^3 \cdot r^2 \cdot \Omega^4) / (5 \cdot c^5 \cdot r^4) \quad [\text{GR quadrupole}]$$

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## 3. Parameters

| Parameter               | Symbol            | Value                   | Unit        |
|-------------------------|-------------------|-------------------------|-------------|
| Mass                    | $M$               | $2.785 \times 10^{30}$  | kg          |
| Radius                  | $r$               | $2.000 \times 10^4$     | m           |
| Initial B-field         | $B_0$             | $1.00 \times 10^{10}$   | T           |
| B-field decay timescale | $\tau_B$          | $1.262 \times 10^{11}$  | s (4000 yr) |
| Critical B-field        | $B_{\text{crit}}$ | $1.00 \times 10^{11}$   | T           |
| Initial spin rate       | $\Omega_0$        | $2\pi/5 \approx 1.2566$ | rad/s       |

| Parameter           | Symbol               | Value                   | Unit            |
|---------------------|----------------------|-------------------------|-----------------|
| Spin-down timescale | $\tau_{\text{spin}}$ | $3.156 \times 10^{11}$  | s (10 kyr)      |
| Hubble constant     | $H_0$                | $2.184 \times 10^{-18}$ | $\text{s}^{-1}$ |

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#### 4. Numerical Result (t = 5000 yr)

$$t = 5000 \times 3.156 \times 10^7 = 1.578 \times 10^{11} \text{ s}$$

$$\begin{aligned} B(t) &= 1 \times 10^{10} \times \exp(-1.578 \times 10^{11} / 1.262 \times 10^{11}) \\ &= 1 \times 10^{10} \times \exp(-1.25) \approx 2.865 \times 10^9 \text{ T} \end{aligned}$$

$$(1 - B/B_{\text{crit}}) = 1 - 2.865 \times 10^9 / 1 \times 10^{11} = 0.97135$$

$$\begin{aligned} \Omega(t) &= 1.2566 \times \exp(-1.578 \times 10^{11} / 3.156 \times 10^{11}) \\ &= 1.2566 \times \exp(-0.5) \approx 0.7616 \text{ rad/s} \end{aligned}$$

$$\begin{aligned} g_{\text{Magnetar}}(t=5\text{kyr}) &\approx (G \cdot M / r^2) \times 0.97135 \times (1 + H_0 \cdot t) \\ &\approx 4.607 \times 10^{11} \times 0.97135 \times (1 + \text{small}) \\ &\quad + 1.007 \times 10^{12} \quad [\text{Ug1+Ug4 floor term}] \\ &\approx 4.474 \times 10^{12} \text{ m/s}^2 \end{aligned}$$


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#### 5. Available Equations for Magnetar Systems

- $g(t)$  — surface gravity evolution (primary)
  - $B(t) = B_0 \cdot \exp(-t/\tau_B)$  — magnetic decay
  - $\Omega(t) = \Omega_0 \cdot \exp(-t/\tau_{\text{spin}})$  — spin-down
  - $P(t) = 2\pi/\Omega(t)$  — pulse period vs time
  - $\Delta P/P = \tau_{\text{GW}} / \tau_{\text{spin}}$  — characteristic age
  - $L_X(t) \propto B(t)^2 \cdot \Omega(t)^4$  — X-ray luminosity proxy
  - $r_s = 2GM/c^2$  — Schwarzschild radius ( $r_s \approx 4.138 \text{ km}$ )
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#### 6. Conclusions

The UQFF magnetar gravity model reproduces  $g \approx 4.474 \times 10^{12} \text{ m/s}^2$  at  $t = 5 \text{ kyr}$  for a canonical  $1.4 M_{\odot}$  magnetar with  $r = 20 \text{ km}$ , consistent with X-ray pulse timing constraints. The magnetic-suppression and Hubble-expansion corrections together account for  $\sim 3\%$  deviations from the static GR prediction. PAPER\_753, CP4 class #337. v5.39.